Neeraj Kumar

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EDUCATION

Northeastern University | *Boston, MA* **Expected May 2027**
Master of Science in Computer Science (Khoury College of Computer Sciences)
Relevant Coursework: Algorithms, Data Structures, Object-Oriented Design, Database Management Systems, Natural Language Processing, Cloud Computing, Human-Computer Interaction

SKILLS

Languages: Python, Java, JavaScript, TypeScript, SQL, C | **Frontend:** React, Next.js, Vite, HTML, CSS, Tailwind CSS, Framer Motion | **Backend and Cloud:** Node.js, REST APIs, AWS (EC2, Lambda, CloudFormation, SNS, SQS), Docker, GitHub Actions, Linux | **Databases:** MySQL, MongoDB | **AI/ML:** OpenAI API, Hugging Face Transformers, local LLM inference (GPT-2), transformer architectures, attention mechanisms (MHA, GQA, MLA), NLP (TF-IDF, Word2Vec, n-gram language models), prompt engineering | **Testing and Tools:** Git, Selenium, JUnit, Figma, Chrome Extension API

WORK EXPERIENCE

Northeastern University | *Graduate Teaching Assistant, Human-Computer Interaction* **Sep 2026 – Present**
- Mentor 6 student teams (25 students) through a semester-long user-centered design project, from problem framing and user interview analysis to low- and high-fidelity prototypes.
- Grade team and individual deliverables against a standardized rubric and give written feedback at each project milestone.

LongevAI Inc. | *AI Developer Intern* **Feb 2026 – May 2026**
- Built 10 role-based dashboards in React and Vite for the company's geriatric care platform using GitHub Copilot, with a reusable dashboard shell that sped up each new dashboard after the first.
- Built a JavaScript UI auditing tool for a client's 5-role veterinary clinic management system and documented 106 bugs, producing the QA report the team adopted as its main remediation reference.
- Redesigned 8 screens and led a full visual rebrand of the client system, applying a unified design token system across the sidebar, header, and dashboards, approved and adopted by the client.

Latitude Health | *Software Engineering Intern* **May 2024 – Aug 2024**
- Delivered a Utilization Management reviewer dashboard in 2 weeks from Figma designs using React and TypeScript.
- Built 9 reusable React components (tooltip, status, priority indicators) adopted across multiple pages.
- Developed 4 user-facing pages, including SSO login and error-handling flows, supporting the company's first client onboarding.

Wyoming Global Research World Pvt. Ltd. | *Web Developer* **Oct 2022 – Nov 2023**
- Integrated a ChatGPT-powered chatbot into educational websites after evaluating 3 frameworks on response latency, usability, and integration complexity.

Cognizant | *Software Intern* **Apr 2022 – Aug 2022**
- Ran automated test suites in Selenium, Java, and XPath to validate UI behavior and frontend-backend data integrity across 4 landing pages as part of a 6-engineer Agile team.

PROJECTS

LLM Explainer | *React, Vite, Framer Motion*
- Built an interactive visualizer of transformer architectures covering 13 LLMs across 6 model families, using a data-driven model registry to render architecture-specific components (attention, activation, positional encoding) with no per-model hardcoding.
- Designed pipeline diagrams that adapt to each model's config, so new models need no core rendering changes.

UI Inspector | *JavaScript, Chrome Extension API, HTML, CSS*
- Rebuilt the LongevAI QA script into a Manifest V3 Chrome extension that traverses the React Fiber tree to map page elements to their source components and captures element screenshots with html2canvas.
- Logged severity-tagged issues with computed styles in a persistent sidebar, with JSON and CSV export for developer handoff.

MassWater | *React, Vite, NOAA API*
- Built a ski conditions tracker using live NOAA API data from 6 stations across MA, NH, and VT, with a 0-100 readiness score weighting snow depth, temperature, and wind speed into a single recommendation.

PUBLICATION

Fuzzy-Based Hierarchical Routing Protocol for Wireless Sensor Networks | *Springer (Sustainable Computing)* **Jan 2023**
- Co-authored a fuzzy-logic routing protocol that doubled network lifetime over standard LEACH (7,000 vs. ~3,500 rounds) and kept 57 of 100 nodes alive at simulation end versus 2, validated in MATLAB.